

Choose the correct answer from the given alternatives

- 1) Fruits and vegetables account about 95 percent vitamin C; vitamin C is water soluble, cannot be stored in the body and have benefits for the body for
- A. Protecting the body from paralyses
 - B. Breast development
 - C. Promoting healthy cell development
 - D. Increase fertility

ANSWER: C

- 2) which is the division of plant physiology dealing with functional processes in plant material after it has been harvested.
- A. Post-harvest physiology
 - B. Plant propagation
 - C. Crop maturity
 - D. All of the above

ANSWER: A

- 3) For post- harvest loss different factors have a contribution; from those which is an external factor for the loss
- A. Biological
 - B. Chemical
 - C. Temperature
 - D. Mechanical

ANSWER: C

- 4) Temperature, concentration of CO_2 and oxygen are the main environmental factors which influence the
- A. Rate of growth of fruit and vegetables
 - B. Rate of respiration
 - C. Yield
 - D. All of the above

ANSWER: B

- 5) Storage life of fruits and vegetables is extended by keeping them in an atmospheric condition which combines
- A. High CO_2 and low in oxygen
 - B. High in oxygen and low in CO_2
 - C. At low temperature
 - D. At high relative humidity

ANSWER: A

- 6) Which is a primary cause for post-harvest loss in horticultural crops and initiated by undesirable reactions between chemical compounds present in the food and
- A. Spoilage
 - B. Biological factors
 - C. Chemical compounds
 - D. Bruising

ANSWER: C

- 7) Chilling injuries arising from the exposure of the products to a temperature
- A. Above the normal physiological range
 - B. Below the normal physiological range
 - C. Under poor ventilation condition
 - D. In CA (controlled atmospheric) storage

ANSWER: B

- 8) Volatile toxic substances are accumulated in the tissue of harvested fruits when they are stored in which storage environment
- A. Above the normal physiological range
 - B. Below the normal physiological range
 - C. Under poor ventilation condition
 - D. In CA (controlled atmospheric) storage

ANSWER: C

- 9) Which control strategy is likely to have the greatest impact on non-target organisms?
- A. Chemical control
 - B. Cultural control
 - C. Biological control
 - D. Physical/mechanical control

ANSWER: A

- 10) Sucking mouthparts are NOT found in:
- A. Fleas
 - B. Lice
 - C. Flies
 - D. Ants

ANSWER: D

- 11) Which developmental stage is causing damage on plants?
- A. Prepupa
 - B. Larvae
 - C. Pupa
 - D. Pupae

ANSWER: B

12) Chemicals which inhibit feeding or piercing

- A. Suppressants
- B. Deterrents
- C. Attractants
- D. Repellents

ANSWER: A

13) Ability of a chemical to bring about changes in the biological system of the target animal

- A. Acute toxicity
- B. Chronic toxicity
- C. Toxicity
- D. Poisoning

ANSWER: B

14) Which of the following is a cultural method for controlling insects?

- A. Crop rotation
- B. Quarantine
- C. Window screen
- D. Sex pheromone traps

ANSWER: A

15) Which of these denotes a type of host plant resistance characterized by the ability of the plant to outgrow and or repair damage resulting from an insect attack?

- A. Antibiosis
- B. Antixenosis
- C. Symbiosis
- D. Tolerance

ANSWER: D

16) Which of the following is correct regarding sexual propagation?

- A. Uniparental
- B. Biparental
- C. Genome is recombination of parents genes
- D. Tri-parents

ANSWER: B

17) Which of the following is not merits of grafting

- A. Create special growth habit
- B. Change a cultivar
- C. Replace tree injuries
- D. Poor healing

ANSWER: D

18) The major methods of sexual propagation is

- A. Cutting
- B. Layering
- C. Division
- D. Seed

ANSWER: D

19) What is acclimatization in the plant propagation?

- A. The process in which the seedling adjusts to a change in its environment
- B. The artificial process of producing plants vegetatively through tissue culture or cell culture techniques
- C. Fast vegetative propagation methods of plants
- D. The act of joining two plants together

ANSWER: A

20) Which of the following is a method for the artificial propagation of a plant?

- A. Germination
- B. Runners
- C. Budding
- D. Seed Sowing

ANSWER: C

21) Which is an example of sexual plant reproduction?

- A. Tissue culture
- B. Grafting
- C. Cuttings
- D. Seed dispersal

ANSWER: D

22) Which of the following is the most producing coffee species world wide?

- A. Coffee arabica
- B. Coffee robusta
- C. Coffea canephora

ANSWER: A

23) The seeds of coffee is known as

- A. Beans
- B. Cherry
- C. Coffee
- D. Fruits

ANSWER: A

24) What is slippery skin of the coffee bean called?

- A. Exocarp
- B. Parchment
- C. Peaberry
- D. Mucilage

ANSWER: D

25) Which of the following is true regarding *coffee arabica*

- A. Susceptible to rust disease

- B. Resistance to rust diseases
- C. Less flavour and acidity
- D. Self incompatible

ANSWER: A

26) Plagiotropic growth habit of coffee is

- A. Verticla or errect growth habit
- B. Horizontls growth habit
- C. Both virtical and horizontal growth habit
- D. Shade loving

ANSWER: B

27) Which of the following is true regarding coffee seed germination

- A. Coffee has hypogeal germination type
- B. Coffee has epigeal germination type
- C. Cotyledones are below the surface the grownd
- D. Cotyledones are not germinating

ANSWER: B

28) Which of the following is not the merits of pruning for coffee production

- A. Shape coffee tree
- B. Allow orthotropic growth habit
- C. Minimize bienial bearing
- D. Remove diseased and dead wood

ANSWER: B

29) What colour is a coffee cherry when ripe?

- A. Green
- B. Black
- C. White
- D. Red

ANSWER: D

30) Bending stem of the young coffee trees in an arc (45-45 degree) by pegging them down to the ground for approximately 12 to 18 months.

- A. Capping
- B. Agobiado
- C. Candelabra
- D. Rehabilitation

ANSWER: B

31) Which of the following is true regarding spices?

- A. Enhance the taste and flavour of food
- B. The aromatic oil obtained by distillation of plant material
- C. Flowering parts of the plant
- D. Fresh parts of the plant

ANSWER: A

32) Olericulture is

- A. Fruits
- B. Ornamental plants
- C. Vegetable and root/tuber crops
- D. Medicinal plants

ANSWER: C

33) What is a plant taxonomy?

- A. A collection of plants that share characteristics grouped together.
- B. Finds, identifies, describes, classifies, and names of plants
- C. A group of plants that share the same features
- D. A class of plants whose members have the same main characteristics

ANSWER: B

34) Which of the following is the genus name of Ginger?

- A. Ginger
- B. Officinale
- C. Zingiber
- D. Coriandrum

ANSWER: C

35) Which of the following is not a method of medicinal plant extraction?

- A. Maceration
- B. Decoction

- C. Percolation
- D. Drying

ANSWER: D

- 36) The plant which is used as a source of spice as well as dye is
- A. Turmeric
 - B. Cardamom
 - C. Clove
 - D. Indigofera

ANSWER: A

- 37) Which of the following is true
- A. Tea is alcoholic beverage
 - B. Tea is a non-alcoholic beverage
 - C. Tea is non-aromatic beverage
 - D. Tea is a drug plant

ANSWER: B

- 38) Why tea is importance
- A. It prevent kidney stones, cold and cough and relieves stress
 - B. Nul antioxidant capacity
 - C. Increase anxiety, stress, and restlessness
 - D. Hurt burn

ANSWER: A

- 39) The major constraints of tea production in Ethiopia
- A. Absence of research facilities to promote tea research
 - B. An increase in demand
 - C. High consumption by neighbouring countries
 - D. Favourable investment policy

ANSWER: A

- 40) Which of the following is the genus name of tea?
- A. Camellia
 - B. Sinensis
 - C. Camellia sinensis
 - D. Tea

ANSWER: A

41) What is epigeal type of tea germination?

- A. Cotyledones remain below the soil surface
- B. Cotyledones remain above the soil surface
- C. Cotyledones remain in the seed coat
- D. Cotyledones remain in the seed

ANSWER: B

42) Which of the following is not describing the stages of growth and development of tea plant?

- A. Seedling/young plant stage
- B. Branch formation stage
- C. Low vigor/degraded bush stage
- D. Bud growing stage

ANSWER: D

43) Which of the following should not be a criterion for a good research project?

- A. Demonstrates the abilities of the researcher
- B. Is dependent on the completion of other projects
- C. Demonstrates the integration of different fields of knowledge
- D. Develops the skills of the researcher

ANSWER: B

44) Research that seeks to examine the findings of a study by using the same design but a different sample is which of the following?

- A. An exploratory study
- B. A replication study
- C. An empirical study
- D. Hypothesis testing

ANSWER: B

45) Which research method is a bottom-up approach to research?

- A. Deductive method
- B. Explanatory method
- C. Inductive method
- D. Exploratory method

ANSWER: C

46) A qualitative research problem statement:

- A. Specifies the research methods to be utilized
- B. Specifies a research hypothesis
- C. Expresses a relationship between variables
- D. Conveys a sense of emerging design

ANSWER: D

47) A variable that is presumed to cause a change in another variable is called:

- A. An intervening variable
- B. A dependent variable
- C. An independent variable
- D. A numerical variable

ANSWER: C

48) In an experiment, the group that does not receive the intervention is called:

- A. The experimental groups
- B. The participant groups
- C. The control groups
- D. The treatment groups

ANSWER: C

49) Which of the following is wrong about the characteristic of root and tuber crops?

- A. Root and tuber crops have a prolonged storability.
- B. Root and tuber crops adapt to a wide range of agro-ecologies.
- C. Root and tuber crops production is labor-intensive duty.
- D. Marketing problems due to high unit weight and rapid decay.

ANSWER: A

50) Which of the following disease is the most serious disease of potato?

- A. Leaf rust
- B. Late blight
- C. Bacterial root rot
- D. Mosaic virus

ANSWER: B

51) Which of the following sentence is true about sweet potato vine cuttings?

- A. Vine cuttings of sweet potato taken from young plants (2-3 months) tend to produce higher yields than vine cuttings taken from old plants (4-5 months)
- B. Sweet potato cuttings from the tips of the vine are not good planting material
- C. Sweet potato cuttings from the base will produce a higher yield
- D. Long vine cuttings tend to produce lower yields than shorter ones

ANSWER: A

52) Which of the following is not the propagation method of potato?

- A. Seed tubers
- B. True potato seed
- C. Stem cuttings
- D. Root cuttings

ANSWER: D

53) Cassava is considered as an ideal food crop in the tropics due to

- A. Its characteristics of resistance to drought (tolerant up to six months)
- B. Its adaptation to high pH soils
- C. Its low carbohydrate yield/unit area
- D. Grows in relatively rich soils

ANSWER: A

54) Which of the following sentence is wrong about anchote production?

- A. Anchote tubers can be kept in the soil for up to two years with no serious loss in quality.
- B. Anchote can only be propagated vegetatively.
- C. Anchote responds strongly to soil fertility, particularly to wood ash.
- D. Anchote belongs to the family cucurbitaceous

ANSWER: B

55) _ is a branch of horticulture which deals with the science or practice of growing plants for aesthetic uses

- A. Olericulture
- B. Ornamental horticulture
- C. Floriculture
- D. Landscape horticulture

ANSWER: B

56) How flower farm employers harvested loose flowers rose for Ethiopian winner athlete's during 2022 world Olympic

- A. With their stalks attached to the flowers
- B. Their leaves
- C. Their roots
- D. Without stalk detached to the flower

ANSWER: D

57) Assume, horticulture graduate form an axon to produce potted plants at **Shewa Robit**, there must be application of A-Rest growth regulator chemical, what is the reason?

- A. To increase height of potted plants
- B. To reduce intermodal length by reducing the number of leaves
- C. To reduce intermodal length without reducing the number of leaves
- D. To decrease the number of flowers

ANSWER: C

58) Which one of the following is **mismatched** about the effects of environmental factors on ornamental plant production

- A. High temperature => small flower bud size
- B. Red light => promote stem elongation
- C. Blue light => improves flower color
- D. Light color => determine flowering of long day plants

ANSWER: D

59) Investors established greenhouses at Holeta largest flower farm, Because Greenhouse

- A. Provides a favorable environment for quality flower production throughout the year
- B. Provide shelter for growing of shade loving plants at summer
- C. Modify environments out of the crop requirements
- D. Is convenient to hardening of planting material solely

ANSWER: A

60) One of the following **is not** the purpose of propagation of ornamental plants

- A. To produce small number of plants in a short period of time
- B. To preserve endemic plants
- C. Employment opportunities for youth
- D. To meet the demand of planting material

ANSWER: A

61) If there is an option, the longer side of nursery bed beds should be oriented from

- A. west to south
- B. east to west
- C. south to north
- D. east to north

ANSWER: B

62) Which is a type of nursery bed often used when the area is known with shortage of rainfall.

- A. raised
- B. flat
- C. sunken
- D. A and B

ANSWER: C

63) Which of the following is a ripening hormone?

- A. Gibberellin
- B. B Auxin
- C. Cytokinin
- D. Ethylene

ANSWER: D

64) During propagation if the propagator wants to initiate roots the ratio of auxin to cytokinin should be

- A. Increased
- B. Decreased
- C. Equal
- D. The ratio has no effect on rooting

ANSWER: A

65) Seeds of mangoes are most convenient for local and international distribution of desirable varieties.

- A. monoembryonic
- B. polyembryonic
- C. Indian

ANSWER: B

66) During shaping or adapting to specific framework of a fruit tree by pruning, tying, and bracing the best framework is

- A. For maximum strength & fruit carrying capacity a scaffold limb should develop with a wide angle at trunk
- B. If a scaffold limb is adjusted to grow vertical and dominate leader shoot
- C. If a scaffold limb adjusted to grow horizontal from the trunk

ANSWER: A

67) The Equilibrium or equality of visual attraction of the landscape is called.

- A. Balance
- B. Rhythm
- C. Form
- D. Focalization

ANSWER: A

68) Which is **NOT** an important principle of landscaping?

- A. Balance
- B. Scale
- C. Form
- D. Harmony

ANSWER: B

69) Which is **NOT** an important element of landscaping?

- A. Color
- B. Style
- C. Line
- D. Unity

ANSWER: D

70) Through repetition of shape, we can create. ?

- A. Balance
- B. Mobility
- C. Form
- D. Rhythm

ANSWER: D

71) Which color creates **nearness**?

- A. Green
- B. Black
- C. Red
- D. Blue

ANSWER: C

72) Site analysis is the

- A. 1st stage of planning
- B. 2nd stage of planning
- C. 3rd stage of planning
- D. 4th stage of planning

ANSWER: A

73) Which is the science and technology that deals with production, storage, processing and marketing and utilization of vegetables.

- A. Olericulture
- B. Olericulturist
- C. Pomology
- D. Floriculture

ANSWER: A

- 74) Which one of the following is a type of gardening which requires agreement between factory owner and farmers?
- A. Market gardening
 - B. Truck gardening
 - C. Processing gardening
 - D. Forcing gardening

ANSWER: C

- 75) Which refers to the period of the natural death of the plants
- A. Ripening
 - B. Senescence
 - C. Maturity
 - D. Harvesting

ANSWER: B

- 76) What are the major soil sterilization methods used in most growers?
- A. Solarization, Fumigation, Steam sterilization and Raw heat method
 - B. Solarization, Fumigation, and Steam sterilization
 - C. Fumigation, Steam sterilization and Raw heat method
 - D. Solarization, Steam sterilization and Raw heat method

ANSWER: A

- 77) When we select the site for vegetable production we should consider?
- A. Field topography, Climate, Water and Soil type
 - B. Only water source
 - C. Soil type only
 - D. Water and Soil type only

ANSWER: A

78) One is different from the other?

- A. Staking
- B. Pruning
- C. Hardening off
- D. Garnish

ANSWER: **D**

79) Garlic is planted from its individual segments, called

- A. Bulb
- B. cloves
- C. rounds
- D. rhizome

ANSWER: **B**

80) Green house is a structure enclosed by glass or plastic that allows light transmission for plant growth

- A. True
- B. False

ANSWER: A

81) Which one of the following is important of green house?

- A. Protection plant from damage
- B. Inefficient application of pesticide in closed environment
- C. Difficulty of maintaining a climate
- D. Prohibit work

ANSWER: A

82) Which one of the following is not advantage of greenhouse benches?

- A. Plants are at a more convenient height
- B. It permit more effective display of plants
- C. It provide improved air circulation
- D. Low quality of product

ANSWER: D

83) Which one of the following is not consider during construction of green bench

- A. Weight capacity
- B. Cost
- C. Air flow
- D. Discontinuity

ANSWER: D

84) Type of green house that consist of two or more individual greenhouse connected together

- A. Gutter connected
- B. Detached greenhouse
- C. Gothic arc
- D. Quonset

ANSWER: A

85) Green house that provides support for the covering

- A. Foundation
- B. Frame
- C. Eave
- D. Ventilator

ANSWER: B

86) Haustoria are produced.

- A. By fungi parasitising animal tissues
- B. By hypha of obligate parasite
- C. In artificial cultures
- D. After the hyphae become intercellular

ANSWER: B

87) Stroma is

- A. Compact somatic hyphae with fruit bodies
- B. Loosely interwoven hyphae
- C. A small hyphae branch
- D. A group of spores

ANSWER: A

88) All fungi are

- A. Autotrophs
- B. Heterotrophs
- C. Saprophytes
- D. Parasites

ANSWER: B

89) Myxomycetous fungi are also called

- A. Slime molds
- B. Bread molds
- C. Black mold
- D. Blue molds

ANSWER: A

90) *Plasmodiophora brassica* causes a disease in crucifers which is known as

- A. Club root
- B. White rust
- C. Early blight
- D. Late blight

ANSWER: A

91) Motile asexual reproductive bodies are called

- A. Chlamydospores
- B. Zoospores
- C. Aplanospores
- D. Zygosporos

ANSWER: B

92) Wart disease of potato is caused by

- A. *Syncytrium endobioticum*
- B. *Alternaria solani*
- C. *Pytophthora infestans*
- D. *Puccinia graminis*

ANSWER: A

93) Haustoria's are produced by

- A. Fungal parasites of animal
- B. intercellular hyphae of obligate parasites
- C. fungi growing on artificial media
- D. intercellular hyphae of parasitic fungi

ANSWER: B

94) Damping off of seedling is caused by

- A. *Albugo candida*
- B. *Pythium debaryanum*
- C. *Sclerospora graminicola*
- D. *Peeronospora parasitica*

ANSWER: B

95) Which fruit production is suitable for Debre Berihan and similar agro- ecology of Ethiopia.

- A. Pear
- B. Banana
- C. Mango
- D. Avocado

ANSWER: A

96) Which parts of Citrus fruit consists of the colored portion of the peel contain the carteniodes, which gives color to the fruit.

- A. Albedo
- B. Flavedo
- C. Mesocarp
- D. Endocarp

ANSWER: B

97) The Opportunities of citrus production in Ethiopia

- A. Research gab
- B. planting material limitation (no organization for certified)
- C. poor management practice
- D. suitable land and water source

ANSWER: D

98) Which one is the Criteria for selection of root stock in avocado **except one**.

- A. Better adapted to specific climate and soil condition
- B. Dwarfing in nature
- C. Resistant to root rot disease
- D. Susiptable to Salinity

ANSWER: D

99) The common way of Avocado propagation is by

- A. Seed
- B. Cutting
- C. Grafting
- D. Sexual propagation

ANSWER: C

100) Disease affects immature fruits and shows up when fruits ripen as small, brown, sunken lesions, which soften 7-15 days after harvest of Avocado is called

- A. Anthracnose
- B. Scab
- C. Root rot
- D. Aphides

ANSWER: A